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COGNITIVE NEUROSCIENCE

Seal and sea lion brains have evolved to support volitional control of vocal behavior and learning

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Seals and sea lions have highly developed volitional breathing control, to which the phocid seals add vocal production learning, including mimicry. In this work, using histology and ex vivo diffusion magnetic resonance imaging tractography, we provide evidence for a phylogenetic spectrum of accumulative neural adaptations supporting aspects of volitional vocal control across pinnipeds. Otariids and phocid seals, but not coyotes, have a direct connection between the vocal motor cortex and phonatory brainstem nuclei. Harbor seals showed hypertrophic connectivity between the anterior ventrolateral thalamus and the vocal premotor cortex—part of a forebrain circuit related to vocal learning in birds and mimicry in humans and parrots. We demonstrate that phocid seals have auditory-premotor pathways potentially related to developmental call learning.

Vocal behavior in most animal species has been considered inflexible or obligate (*1*) and is governed by interactions between an intricate network of midbrain and brainstem nuclei (*2*–*4*) that predominantly respond to environmental and affective conditions. Nonobligate vocal behavior—the product of vocal learning—is increasingly viewed as a spectrum of interrelated capabilities (*5*) with increased cognitive control of respiratory and phonatory motor systems required for the most vocally flexible species. A growing number of avian and mammalian species have been demonstrated to deploy species-typical calls in novel contexts [referred to as usage learning (*1*, *6*, *7*)]. Much rarer are species whose members can learn to produce new calls with frequency and formant characteristics outside of the inherited repertoire. These vocal production learners range from those who can acquire species-specific call types through learning to those who can floridly mimic sounds well outside their natural repertoire. Production learning is most common in avian lineages, where the behavioral and neurobiological mechanisms of vocal learning have been extensively studied in both the field and laboratory (*8*). Such studies have indicated that the ability to alter both filter (formants, via the entire vocal tract) and source phonation (frequency, via the syrinx) components of species-typical calls relies on the elaboration or doubling of pathways in two forebrain circuits (*9*). The posterior circuit directly connects the vocal motor cortex with brainstem phonatory neurons, allowing volitional control of the syrinx and vocal tract. The anterior circuit connects vocal premotor regions with the anterior striatum and anterior ventrolateral thalamus (*10*) and facilitates vocal motor reinforcement learning.

The study of vocal learning in birds has exemplified the benefit of a model clade (*11*) approach to comparative behavioral neuroscience. Because scientists have been able to examine closely related bird species that vary in systematic ways in terms of vocal flexibility (e.g., oscine and suboscine songbirds and parrots), the neurobiological underpinnings of different aspects of call production and learning in birds have been elucidated. Comparisons of primate and avian vocal neurobiology have been productive and have led to a number of hypotheses about neurobiological adaptations supporting vocal plasticity and learning (*12*, *13*). Humans may be unique among primates in their sharing an analog to the avian posterior pathway connecting the vocal motor cortex to brainstem phonatory nuclei (*14*, *15*). The Kuypers-Jürgens hypothesis (*16*) states that these direct synaptic connections are an evolutionary precondition for vocal production flexibility. Humans also show enhanced connectivity in striatal thalamocortical circuits analogous to the avian anterior pathway (*12*, *17*). Although not emphasized in studies of bird vocal learning, humans have increased auditory-premotor connectivity (e.g., the arcuate fasciculus connecting the Wernicke's and Broca's areas) that supports developmental language learning (*18*, *19*) and is believed to be relevant to the evolution of human speech (*20*, *21*). Within-clade analyses of mammalian brain circuits related to graded vocal flexibility would further illuminate brain-behavior relationships in mammalian vocal learning.

The behavioral case for graded vocal flexibility in pinnipeds

Species in the suborder Pinnipedia (hereafter referred to as pinnipeds) show clear behavioral evidence of differentially distributed components of vocal flexibility, ranging from flexible usage learning through developmental production learning of species-specific calls and, potentially, mimicry (*22*, *23*). As such, they are a promising model clade for exploring the full spectrum of vocal learning adaptations in the mammalian brain (*1*, *5*, *6*, *24*). Pinnipeds are a lineage of marine-adapted Carnivores comprising odobenids (family Odobenidae, walruses), otariids (family Otarridae, eared seals and sea lions), and phocids (family Phocidae, true seals). They have diverged and radiated over the past 25 million years from a single ancestral amphibious carnivore ancestor related to species within the family Canidae (*25*, *26*). Compared with terrestrial carnivores, pinnipeds have experienced strong adaptive pressures to control breathing, swallowing, and sound production for a semi-aquatic lifestyle. Carnivores, including pinnipeds' closest relatives in the family Mustelidae (mustelids), have not been found to express high vocal production flexibility (*1*). By contrast, all extant pinnipeds show some evidence of enhanced vocal flexibility, if only in the ubiquity and apparent ease of fine-grained, rapid usage learning (*22*).

Extensive fieldwork has shown that otariids produce highly stereotyped calls, with no clear evidence of production learning (*27*). However, there is experimental evidence that otariids can easily learn to produce and inhibit species-typical calls under contextual control (*28*). This appears to contrast with most terrestrial mammals, which, although capable of some degree of usage learning, seem to require extensive training to achieve fairly limited control of call timing and amplitude (*29*), relying, to some extent, on conditioned motivational and affective factors (*30*). The Otariidae and Phocidae branches of pinnipeds split at least 24 million years ago (Ma), and unlike the case in otariids, a number of phocid species, including both subfamilies Monachinae (elephant and monk seals) and Phocinae (including harbor and gray seals), show some evidence of species-specific call production learning over development (*8*, *22*, *31*–*34*). This type of learning involves developmental modulation of phonation based on auditory input and, in birds, requires cortical-brainstem phonatory connections and striatal thalamocortical adaptations (*35*). In humans, developmental vocal learning also relies on the arcuate fasciculus connection between the auditory and premotor regions mentioned above (*18*, *19*).

At the far end of the pinniped flexibility spectrum is the Phocinae branch of the phocid family, which split from the Monachinae more than 15 Ma. Some phocids show evidence of highly flexible vocal production learning and mimicry, which may not be fully constrained to a strict developmental period or a specific species repertoire. Evidence of mimicry has been collected in two harbor seals raised in captivity (*36*–*38*), who learned to replicate human words and phrases. Two other harbor seals in captivity have learned to produce novel formant patterns

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through operant techniques (*39*, *40*). In an experimental context, gray seals learned to mimic new tonal sequences and vowellike formants (*41*). This behavior requires fine-grained volitional control of the parts of the vocal apparatus responsible for filter (mouth, formants) and phonation (larynx, frequency) control. Finally, among the pinnipeds, odobenids (walruses), which are more closely related to the otariids than the phocids, have produced some of the most compelling experimental evidence of vocal production flexibility (*42*, *43*). Across the pinnipeds, this range of capabilities, in conjunction with comparisons with terrestrial carnivore cousins, spans the spectrum from relatively fixed vocal stereotypy through mimicry in a set of closely related species with established phylogenetic branch points. Prior comparative empirical and theoretical work suggests that such abilities rely on graded cognitive control of vocal production systems (*8*). To the extent that this is the case, there should be clear differences in neurobiological organization of vocal production and control systems across these species. Determining covariance between neurobiological and behavioral traits across the pinnipeds will thus yield insight into how changes to mammalian brains support vocal flexibility and learning.

In this study, we directly assessed neural pathways potentially related to vocal plasticity and learning in several pinniped species. Pinnipeds are in the canid branch of order Carnivora, so to examine whether their neurobiological adaptations may have emerged after adaptation to the aquatic environment, we compared these pathways with those in a relatively large-brained and fully terrestrial carnivore, the coyote. This allowed us to assemble a preliminary neurobehavioral phylogeny of vocal learning in the canid carnivore line. We used a postmortem-optimized diffusion imaging sequence (*44*–*46*) to scan opportunistically obtained brains of four California sea lions (*Zalophus californianus*, Otariidae), three northern elephant seals (*Mirounga angustirostris*, Monachinae), four harbor seals (*Phoca vitulina*, Phocinae), and four coyotes (*Canis latrans*, Canidae) (*47*) (subject details are provided in table S1). We then conducted multiregion probabilistic tractography in the right and left hemispheres of each subject of the following pathways (table S2). First, we examined connections between the nucleus ambiguus and vocal motor cortex (often referred to as the laryngeal motor cortex) to seek support for the Kuypers-Jürgens hypothesis. Next, we examined connections between the anterior ventrolateral thalamus, anterior striatum, and pre–vocal motor cortex, analogous to the anterior forebrain pathway in birds. To look for an arcuate-like connection potentially related to developmental vocal learning, we examined auditory cortex connections to the pre–vocal motor cortex. Finally, we assessed connections between the anterior cingulate and periaqueductal gray (PAG), associated with volitional deployment of species-typical calls in primates (*48*), To establish seeds and regions for probabilistic tracing, we manually segmented all regions (25 per subject, 375 total; Fig. 1 and fig. S1) using multiple methodologies.

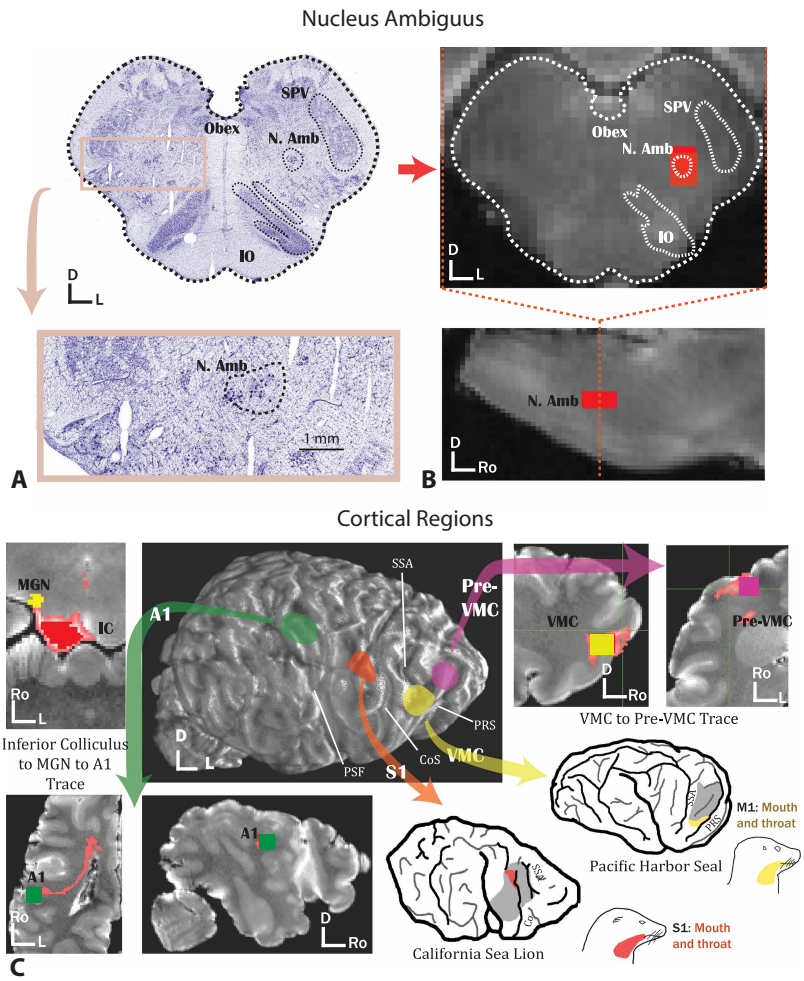

Fig. 1. Location and identification methodology for brain regions. (**A**) Cresyl violet Nissl stain of the brainstem of a California sea lion (CSL) in an axial section, localizing the nucleus ambiguus (N. Amb) in the brainstem with a magnified view below. IO, inferior olive; SPV, spinal trigeminal nucleus; D, dorsal; L, left. (**B**) Histology slices were overlaid over high-resolution MR images of each species based on anatomical landmarks, and all voxels overlapping with the nucleus ambiguus were included in MR segmentations. A brainstem of a CSL is shown in axial (above) and sagittal (below) planes. Ro, rostral. (**C**) Three-dimensional (3D) rendering of a CSL brain with cortical regions of interest (ROIs) overlaid and validations of auditory (green), sensory (orange), motor (yellow), and vocal motor cortex (VMC) and pre-VMC (purple) surrounding. (Left) A1 was identified using direct probabilistic traces from the medial geniculate nucleus (MGN), identified anatomically and validated by tracing from the inferior colliculus (IC). (Top right) The pre-VMC was identified by probabilistic tracing from the VMC and validated by comparison with electrophysiological studies in terrestrial carnivores (ferret and dog). (Bottom right) Representation of the somatosensory cortex in gray, with the mouth and neck cortex in red in the CSL and the primary motor cortex in gray with the VMC in yellow for the harbor seal. These regions were identified with histology and invasive electrophysiology (*66*, *67*). [Figures adapted by H. Cook] Identifications of subcortical regions are provided in fig. S1. PSF, pseudosylivan fissure; PRS, presylvian sulcus; CoS, coronal sulcus; SSA, suprasylvian sulcus.

Results

We found clear evidence of brain circuit differences potentially corresponding to the current differential behavioral evidence for vocal flexibility and learning in each species. There was robust, bilateral, and apparently direct cortical connectivity between the putative vocal motor cortex and the nucleus ambiguus in the brainstem in all pinnipeds and in none of the coyotes (Fig. 2). Notably, there were no significant differences between the coyotes and any of the other species in the two control tracts—the whole brainstem and the corticospinal pathway (fig. S2). Elephant seals and harbor seals, but not sea lions, showed robust arcuate-like connectivity between the putative auditory cortex and vocal premotor cortex. The harbor seal brains showed exceptionally robust anterior cingulate–to–PAG connectivity compared with all other brains evaluated. Finally, harbor seals showed enhanced connectivity between the anterior ventrolateral thalamus and the pre–vocal motor cortex, one pathway in the putative forebrain anterior vocal production learning circuit, and one that has been implicated in mimicry in birds (*49*) (Fig. 3 and fig. S3).

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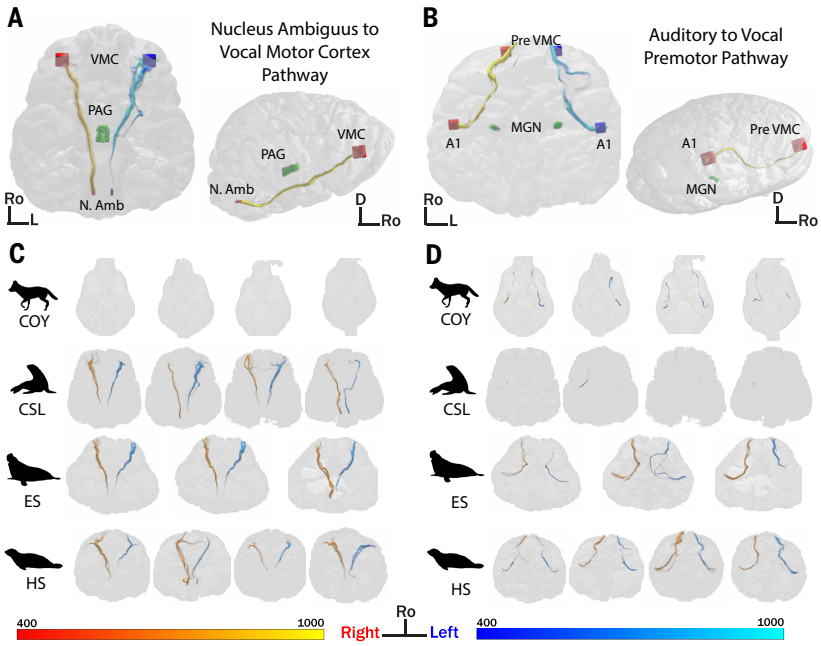

Fig. 2. 3D renderings of left and right nucleus ambiguus–to–VMC and A1–to–pre-VMC pathways for all subjects. Pathways are from probabilistic traces conducted in FSL's PROBTRACKX using standard settings, with a minimum threshold of 400 streamlines. (**A**) Nucleus ambiguus–to–VMC pathways (right, red; left, blue) (excluding the PAG) shown in a CSL brain in the dorsal (left) and sagittal (right) view (movie S1). (**B**) A1–to–pre-VMC pathways (excluding the MGN) shown in a harbor seal brain in the dorsal (left) and sagittal (right) views. (**C**) Results for nucleus ambiguus–to–VMC tracings in all subjects [top to bottom: coyote (COY), CSL, northern elephant seal (ES), and harbor seal (HS)], with a minimum threshold of 400 streamlines. (**D**) Results for A1–to–pre-VMC tracings for all subjects.

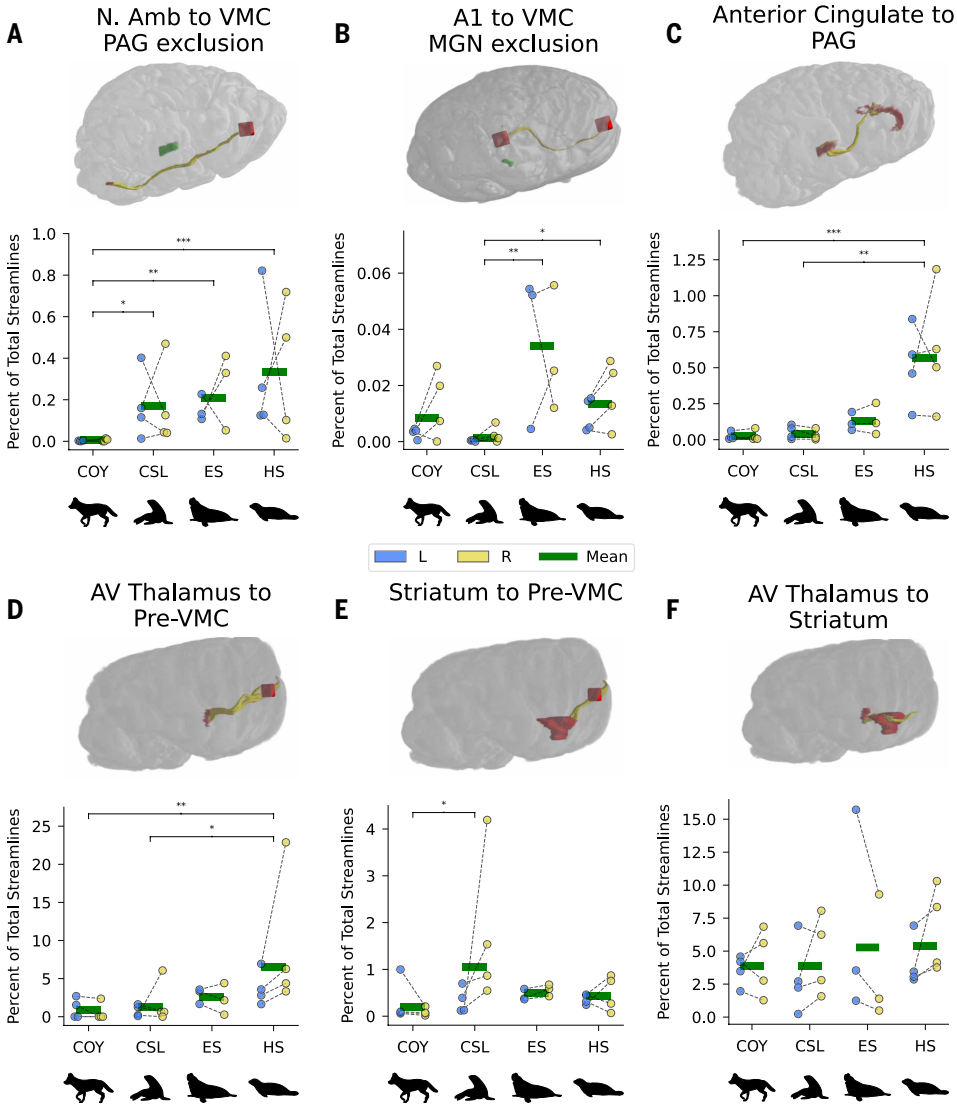

Fig. 3. Corrected tract strength for multiregion probabilistic traces. (**A** to **F**) The six vocal production–relevant pathways are shown in the sagittal view, with seeds and waypoint ROIs in red, exclusion masks (where present) in green, and paths in yellow. Brainstem and corticospinal control tracts and comparisons are provided in fig. S3. Each plot shows the percent of total streamlines for paths on the *y* axis and the species on the *x* axis (tables S1 and S4). Each subject's corrected way-total for each trace is shown for the left hemisphere in blue and the right hemisphere in yellow, with a dotted line connecting the values for each subject. Significant differences between species values are indicated by brackets and asterisks (Kruskal-Wallis test with Bonferroni-corrected pairwise comparisons; $*P < 0.05$; $**P < 0.01$; $***P < 0.001$) (table S5). (A) Nucleus ambiguus–to–VMC traces. (B) A1-to-VMC traces. (C) Anterior cingulate–to–PAG traces. (D) Anterior ventrolateral (AV) thalamus–to–pre-VMC traces. (E) Striatum–to–pre-VMC traces. (F) AV thalamus–to–striatum traces.

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Discussion

Our results demonstrate the existence of a neurobehavioral phylogeny of vocal flexibility in closely related mammalian species with highly variable vocal capability (Fig. 4). One parsimonious interpretation of our findings suggests that an early ancestral pinniped evolved a direct cortical pathway between the vocal motor cortex and phonatory brainstem nuclei. This may have supported robust respiratory control and enhanced usage learning and preceded adaptations supporting vocal production learning that emerged selectively in other pinniped branches. Enhanced auditory-motor cortical pathways supporting developmental vocal production learning may have emerged in the Phocidae after the split from the Otariidae >24 Ma. Robust connections between the vocal premotor cortex and the anterior ventrolateral thalamus as well as enhanced anterior cingulate–to–PAG connections, both potentially related to more robust and cognitively demanding vocal production learning throughout the lifespan, may have emerged in Phocinae after the split with Monachinae 15 Ma. Our findings support, but cannot confirm, the Kuypers-Jürgens hypothesis (*16*). They also emphasize the importance of direct cortical auditory-motor pathways to vocal learning in mammals. Our results do not as clearly implicate anterior striatal thalamocortical circuitry in mammalian vocal plasticity as has been shown in avians, but they do underline the potential contribution of the anterior ventrolateral thalamus–pre–vocal motor cortex pathway to vocal mimicry.

Vocal motor cortex–to–nucleus ambiguus pathway

All pinnipeds tested showed robust and apparently direct connections between cortical vocal motor regions and the nucleus ambiguus without transit through the PAG, which interposes between these regions in studied non–production-learning mammals. The Kuypers-Jürgens hypothesis posits the necessity of a monosynaptic pathway between these regions for vocal production learning. Our findings support this hypothesis, but diffusion tractography cannot confirm whether pathways are monosynaptic or not. The voxels that constrain diffusion tractography are too large to resolve single synapses, and evidence suggests that the technique, although an accurate representation of large myelinated pathways, is not bound by individual synapses (*50*).

Behaviorally, compared with terrestrial carnivores, pinnipeds do show increased vocal motor control. However, there are no data of vocal production learning in the Otariidae. The nucleus ambiguus is part of a network of medullary nuclei controlling the larynx, mouth, and chest musculature involved in breathing and swallowing (*51*, *52*). In most terrestrial mammals, breathing is largely obligate based on peripheral physiological signals, such as blood-gas concentration and lung tension (*53*). When adapting for aquatic life, pinnipeds would have faced intense pressure to develop precise volitional breath control during submersion. Marine mammals are subject to extended apneas during general cortical anesthesia (*54*). Additionally, gray seals (a close relative of harbor seals) have recently been shown to have cognitive control of breathing related to dive behavior (*55*).

We suggest that in pinnipeds, a vocal motor cortex–ambiguus pathway evolved initially to bring vocal timing, breathing, and potentially swallowing under precise volitional control [as suggested by Ravignani *et al.* (*23*)]. This, in turn, would have made control of laryngeal muscles and thoracic muscles involved in phonation more directly accessible to broadly conserved corticostriatal and corticocerebellar general motor learning mechanisms, supporting rapid, precise, and flexible usage learning as observed in the otariids. Further neural adaptations may have been required to support the robust production learning observed in phocid seals.

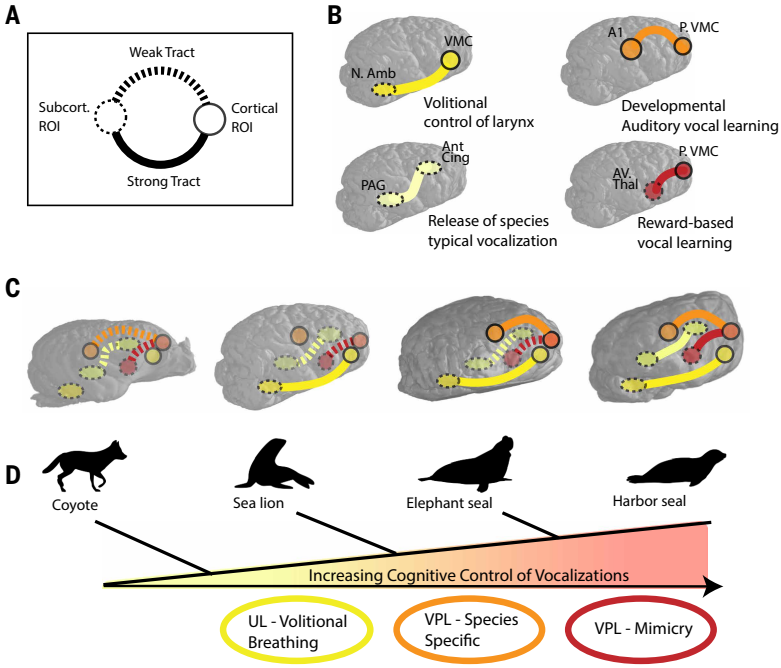

Fig. 4. Tentative neurobiological phylogeny of vocal flexibility in pinnipeds. (**A**) Key showing representation of weak and strong connections (relative to other brains in this study) between cortical and subcortical ROIs. (**B**) Schematic representations of four distinct brain pathways traced in this study that showed significant difference between species and likely relate to vocal flexibility. (**C**) Schematic representations of mean path strength for each species in this study. (**D**) Simplified pinniped phylogenetic tree, representing branch points for the split between the Arctoidae (in which branch the pinnipeds are found) and the Canidae (46 Ma), the Otariidae and Phocidae (>24 Ma), and the Monachinae and Phocinae (10 to 15 Ma). Potential emergence points for different vocal capabilities are indicated below by color, with usage learning (UL) in yellow and vocal production learning (VPL) in orange and red. In line with prior work framing vocal learning capability as a spectrum (*8*), cognitive demand of behavior is represented by a spectrum ranging from limited breath control up through mimicry.

Auditory cortex–to–vocal premotor pathway

Phocids in both the Monachinae and Phocinae lines, but not otariids, show some evidence of vocal production learning and flexibility over development (*56*, *57*). The auditory-motor circuit described in phocids in this study may have emerged or been strengthened in an early phocid to support ecologically relevant developmental vocal learning. Notably, most phocids rely far more heavily on vocal signaling in breeding compared with otariids, which suggests parallel evolutionary pressures to those experienced by songbirds (*58*). The arcuate-like pathway described in this work was notably stronger bilaterally in the two male harbor seal brains compared with the two female harbor seal brains, mirroring findings from seminal bird studies (*59*). We do not yet have clear evidence of production learning in northern elephant seals, but southern elephant seals (*Mirounga leonina*) may use learned vocal signals in breeding contexts (*32*). Notably, despite our limited finding in coyotes, there is evidence of a robust, left-lateralized auditory-premotor pathway in domestic dogs (*60*). Coyotes are not known to be vocal production learners but do chorus socially (*61*).

Anterior striatal thalamocortical circuit

Despite its centrality in neurobiological accounts of vocal learning in birds, the anterior striatal thalamocortical circuit showed limited interspecies differences in this study. The cortical vocal motor–to–brainstem phonatory and the auditory-premotor pathways may be more relevant to vocal plasticity in the pinniped line. Notably, harbor seals did show increased connectivity between the anterior ventrolateral thalamus and the pre–vocal motor cortex. The ventrolateral thalamus in humans, but not in other primates,

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is essential for vocal production (*14*), serving as a hub connecting striatal and cortical regions involved in vocal release and inhibition, learning, and vocal working memory. Thalamic-premotor connections facilitate the transition between call subcomponents in finches (*62*), and elaboration of these circuits in parrots appears to be related to their capability for vocal mimicry throughout the lifespan (*49*). Given this, and the current findings, the anterior ventrolateral thalamus–pre–vocal motor cortex pathway may play a particular role in vocal mimicry in Phocinae seals.

Limitations

Our study's limitations were primarily attributable to sample acquisition—more brains should be collected and analyzed, balanced for age and sex. Lack of common neuroanatomical atlases for these species and limited data on functional division of the cortex are further limitations—these were overcome in the current study through manual segmentation based on multiple lines of prior evidence and anatomical cross-validation. The nucleus ambiguus has been considered particularly difficult to find in magnetic resonance (MR) datasets. Our localization was based on clear, species-specific histology and was further validated by connectivity (table S3 and fig. S4). Notably, we found no evidence of vocal motor cortex projections in the vicinity of the putative nucleus ambiguus in coyotes (fig. S5).

Conclusions

Our study partially validates and qualifies emerging behavioral evidence of vocal plasticity and learning in mammals and complements recent research into vocal production learning in bats (*63–65*). Our findings support the importance of direct cortical motor–to–brainstem phonatory connections for vocal production learning. They further suggest that such connections may be necessary, but not sufficient, for the emergence of highly flexible vocal behavior. Differential cognitive demands of different aspects of vocal production plasticity likely rely on brain circuits that evolve under separate pressures across related species.

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SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.adx9367
Materials and Methods; Figs. S1 to S9; Tables S1 to S5; References (*69–83*); MDAR Reproducibility Checklist; Movie S1

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Seal and sea lion brains have evolved to support volitional control of vocal behavior and learning

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Editor's summary
Humans are vocal learners, but many other species also display this trait, including birds, bats, and seals. Seals within the family Phocidae especially are known for this, and harbor seals are even known to imitate human speech. Cook *et al*. looked at opportunistically obtained brains from seals and sea lions, with coyotes as an outgroup carnivore control. They found a series of neural adaptations that support the volitional control of vocalizations in the marine mammals but not the coyotes, with the harbor seals showing a forebrain circuit similar to that found in mimicking birds and humans.
—Sacha Vignieri

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